

Graphs of functions - Set 1 - Solutions

Specification MM8.1, MM8.5-MM8.7, M4.15.

Answer key

Question	Answer	Topic
1	A	Graphs of functions
2	A	Graphs of functions
3	A	Graphs of functions
4	H	Graphs of functions
5	A	Graphs of functions
6	A	Graphs of functions
7	A	Graphs of functions
8	E	Graphs of functions
9	E	Graphs of functions
10	F	Graphs of functions
11	C	Graphs of functions
12	D	Graphs of functions

Worked solutions

Question 1 GRA-01 · Graphs of functions · spec MM8.6

How many times does the graph of $y = x^3 - x^2 - 6x$ cross or touch the x -axis?

Answer: A

$x(x^2 - x - 6) = x(x - 3)(x + 2)$; three distinct real roots, so three crossings.

Question 2 GRA-02 · Graphs of functions · spec MM8.7

The graphs of $y = x^2 - 2x$ and $y = k$ meet at exactly two points. Find the complete set of values of k .

Answer: A

$y = (x - 1)^2 - 1$ has minimum -1 ; a horizontal line above the minimum meets the parabola twice.

Question 3 GRA-03 · Graphs of functions · spec MM8.1

Which of the following could be the equation of a cubic curve that touches the x -axis at $x = 2$ and crosses it at $x = -1$, and has a negative y -intercept?

Answer: A

Touching at 2 needs the factor $(x - 2)^2$ and crossing at -1 the factor $(x + 1)$. $y = (x - 2)^2(x + 1)$ has y -intercept $4 > 0$, so the sign is reversed: $y = -(x - 2)^2(x + 1)$ with intercept -4 .

Question 4 GRA-05 · Graphs of functions · spec MM8.1

Which of the following statements about the graph of $y = |2x - 4|$ is/are true?

- I It meets the x -axis at exactly one point.
- II It is symmetric about the line $x = 2$.
- III It lies entirely on or above the line $y = 0$.

Answer: H

$|2x - 4| = 0$ only at $x = 2$ (I). $|2x - 4| = 2|x - 2|$ is symmetric about $x = 2$ (II). A modulus is never negative (III).

Question 5 GRA-06 · Graphs of functions · spec MM8.6

The curve $y = x^4 - 5x^2 + 4$ meets the coordinate axes at how many points in total?

Answer: A

x -axis: $(x^2 - 1)(x^2 - 4) = 0$, four points. y -axis: $(0, 4)$, one point. Total 5.

Question 6 GRA-09 · Graphs of functions · spec MM8.1

The graph of $y = \log_2 x$ is reflected in the line $y = x$. The resulting graph has equation

Answer: A

Reflection in $y = x$ gives the inverse function: $x = \log_2 y$, i.e. $y = 2^x$.

Question 7 GRA-10 · Graphs of functions · spec MM8.6

A polynomial of degree 5 with real coefficients has how many real roots at least, and at most?

Answer: A

An odd-degree polynomial tends to opposite infinities at each end, so it crosses the axis at least once; a degree-5 polynomial has at most 5 roots.

Question 8 GRA-12 · Graphs of functions · spec MM8.1

Which of the following functions is undefined at exactly two real values of x ?

Answer: E

$x^2 - 4 = 0$ at $x = \pm 2$: two values. $x^2 + 4$ is never zero; $\frac{x}{x^2}$ and $\frac{1}{(x-2)^2}$ are undefined at one value; $\sqrt{x-2}$ is undefined for all $x < 2$.

Question 9 GRA-16 · Graphs of functions · spec MM8.7

The line $y = mx$ meets the curve $y = x^3 - x$ at three distinct points. Find the complete set of values of m .

Answer: E

$x^3 - x - mx = x(x^2 - (1 + m)) = 0$: three distinct roots iff $1 + m > 0$, i.e. $m > -1$.

Question 10 GRA-04 · Graphs of functions · spec MM8.7

The line $y = x + c$ and the curve $y = \frac{1}{x}$ have no points of intersection. Find the complete set of values of c .

Answer: F

$x^2 + cx - 1 = 0$ always has discriminant $c^2 + 4 > 0$: two intersections for every c . So there are no values of c with no intersection.

Question 11 GRA-07 · Graphs of functions · spec MM8.5

The graph of $y = f(x)$ has $f'(x) > 0$ for $x < 1$, $f'(1) = 0$, $f'(x) > 0$ for $1 < x < 3$, $f'(x) < 0$ for $x > 3$. Which of the following is true?

Answer: C

At $x = 1$ the derivative is zero but does not change sign: a stationary point that is not a turning point (an inflexion).
At $x = 3$ the derivative changes from positive to negative: local maximum.

Question 12 GRA-08 · Graphs of functions · spec MM8.7

How many real solutions does the equation $x^3 = 3x + 1$ have?

Answer: D

Sketch $y = x^3 - 3x - 1$: turning points at $x = \pm 1$ with $y(-1) = 1 > 0$ and $y(1) = -3 < 0$. A cubic whose local maximum is positive and local minimum is negative crosses the axis three times.